

GATE

ALL BRANCHES



General Aptitude

ANALYTICAL APTITUDE

Lecture No.- 03



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Recap of Previous Lecture



Topic 1

Coding Decoding



Topics to be Covered



Topic-1

Directions

Topic-2

Ranking & Arrangements



[MCQ]



#Q. A man went out for evening walk in between 5pm to 6pm and was back home in between 6pm to 7pm. After reaching home, he observed that minute hand and hour hand has interchanged their positions. At what time did the man went out for walk?

A $5:32 \frac{8}{11}$

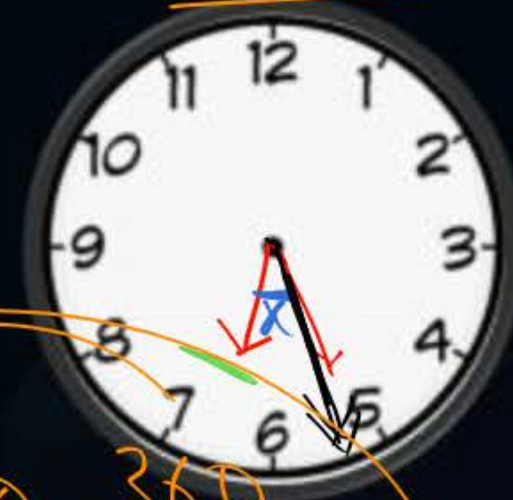
B $5:30$

C $5:32 \frac{4}{13}$

D $5:33$



$150 + x = 180$



$$\frac{180 - 360}{13} = 5 - 5$$

[Sol]



$$\frac{150 + \frac{360}{13}}{5.5}$$

$$H \cdot H \frac{1}{2}^\circ \rightarrow M \cdot H \cdot 6^\circ$$

$$H \cdot H \underline{\underline{1}}^\circ \rightarrow M \cdot H \cdot 12^\circ$$

$$= \frac{1950 + 360}{13} \times \frac{1}{5.5}$$

$$\checkmark \boxed{H \cdot H x^\circ} \rightarrow \boxed{M \cdot H \cdot \underline{\underline{12x^\circ}}}$$

$$= \frac{\overset{210}{\cancel{2310}}}{13} \times \frac{2}{\cancel{11}}$$

$$\checkmark \boxed{360 - x = 12x}$$

$$= \frac{420}{13} = \underline{\underline{32 \frac{4}{13} \text{ numbers}}} \Rightarrow 360 = 13x$$

$$\therefore x = \underline{\underline{\frac{360}{13}}} \checkmark$$

[MCQ]



#Q. In a certain code MAN is written as SANM and WORD is written as SORDW.
How would SALE be written in that code?

Assignment

- A** SEALS
- B** LESAS
- C** LEASS
- D** SALES



[MCQ]



#Q. If CURE is written as BTQD. How will MAIL be written in that code?

Assignment

A LZGS

B LZKH

C LZHK

D LYHK

C U R E → B T Q D

(-1) M A I L → L Z H K

[MCQ]



#Q. If in a certain code TWENTY is written as 863985 and ELEVEN is written as 323039, how is TWELVE written in that code?

Assignment

TWENTY $\rightarrow$ 863985

ELEVEN $\rightarrow$ 323039

TWELVE $\rightarrow$ 863203

A

863203

B

823230

C

823032

D

863903

[MCQ]



#Q. If in a certain language PRIVATE is coded as 1234567 and RISK is coded as 2398, how is RIVETS coded in that language?

Assignment

234769

- A** 687543
- B** 496321
- C** 246598
- D** 234769

[MCQ]



#Q. If 'col tip mot' means 'singing is appreciable'; 'mot baj min' means 'dancing is good' and 'tip nop baj' means 'singing and dancing'. Which of the following means 'good' is that code language?

Assignment

A

min

B

baj

C

nop

D

None of these

mot $\rightarrow$ is

tip $\rightarrow$ singing

col $\rightarrow$ appreciable

baj $\rightarrow$ dancing

good $\rightarrow$ min

[MCQ]



#Q. If light means red, red means green, green means white, white means yellow. What is the colour of blood?

Assignment

A

light

B

blue

C

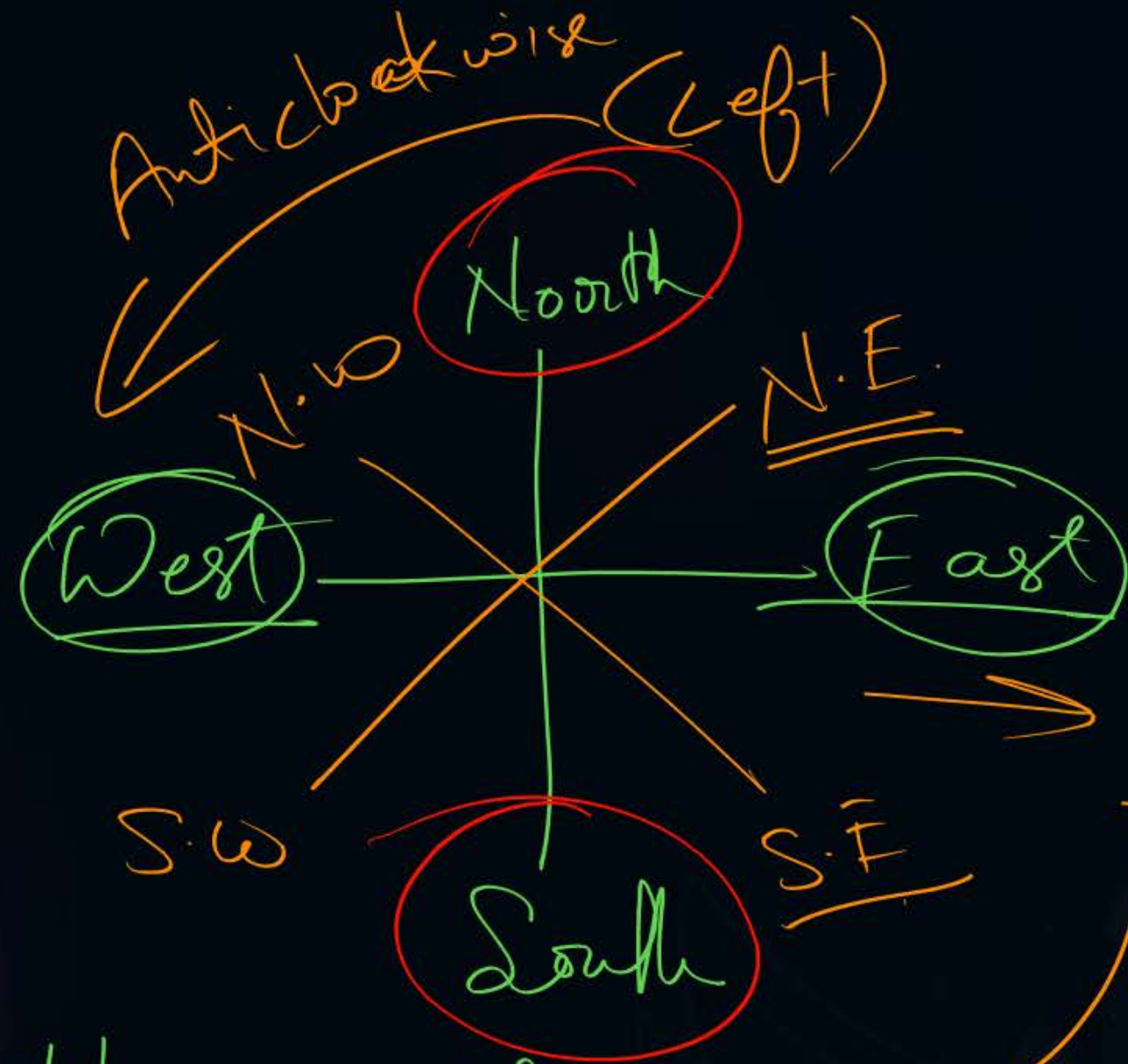
red

D

yellow

Direction

Right? } $\approx 90^\circ$
 Left? }



Vertical
Direction

Horizontal
Direction



Direction

✓
? Displacement

1. Horizontal DISTANCE

2. Vertical

3. Direction from at least

4. Direction as compared from origin
(Left)

5. How far from origin.

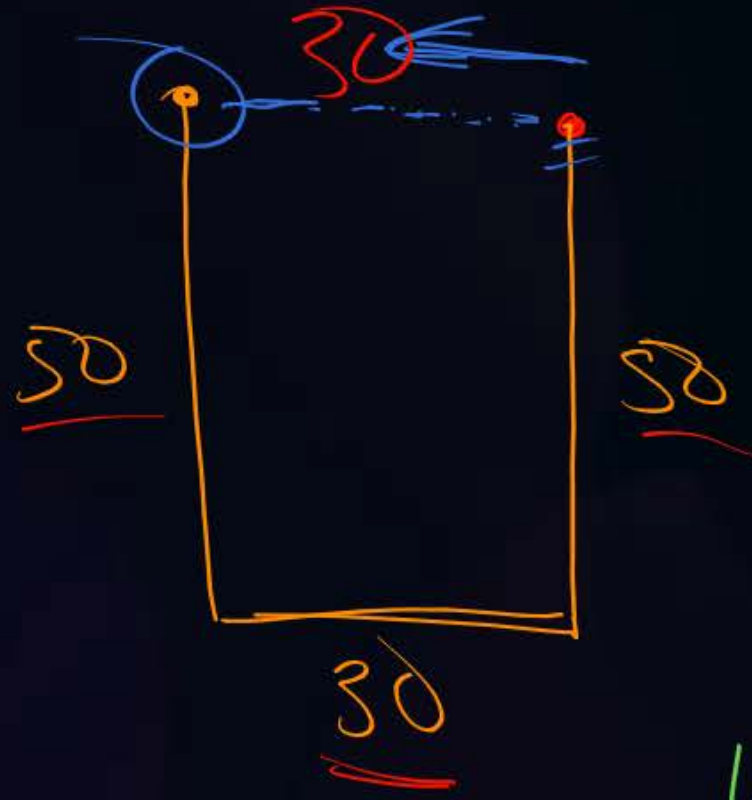


[MCQ]



$$\sqrt{H^2 + V^2}$$

#Q. Facing towards south Suresh walked 50 mts. He then turned to his right and walked 30 mts. He again turned right and walked 50 mts. How far was he from his original position and towards which direction?



$$\sqrt{30^2}$$

30 West
~~S 50~~ W 30 ~~N 50~~



1. West 30
2. 0
3. North

4. West 30
5. 30

[MCQ]



#Q. Mahesh started walking towards North. After walking 30 metres, he turned towards left and walked 40 metres. He then turned left and walked 50 metres. How far is he from his original position?

$$\sqrt{H^2 + V^2}$$

$$= \sqrt{40^2 + 20^2}$$

$$= \sqrt{1600 + 400}$$

$$= \sqrt{2000}$$

$$= \sqrt{4 \times 5 \times 100}$$

$$= 2 \times 10 \sqrt{5}$$

$$= 20\sqrt{5}$$

N₃₀ W₄₀ S₅₀

Vertical $\rightarrow$ 20 South

Horizontal $\rightarrow$ 40 West



44.72m

[MCQ]



#Q. Mohit started from point X and walked straight 5 km West, then turned left and walked straight 2 km and again turned left and walked straight 7 km. In which direction is he from the point X?

South East

W 5 - S 2 - E 7

H $\Rightarrow$ 2 (East)
V $\Rightarrow$ 2 (South)



[MCQ]



#Q. A is standing at P. He walks 10 meters towards the south; then he walks 20 meters towards the west; then he walks 10 meters towards the south; then he walks 20 meters towards the North and reaches Q. What is the straight distance between P and Q in meters?

Assignment

[MCQ]

✓✓ Diagram u



#Q. Sumit walks 8 km east, turns South-West and walks another 8km. He again takes a turn towards North-West and walks another 8km. In which direction from starting point is he standing now?

Assignment

[MCQ]



#Q. A walks 10 metres towards East and then 10 metres to his right, then every time turning to his left, he walks 5.15 and 15 metres respectively. How far is he now from his starting point?

Assignment

- A** 16.55
- B** 6.965
- C** 15.95
- D** 15.0

[MCQ]



#Q. A man is facing North-West. He turns 90° in the clock wise direction, then 135° in the anticlockwise direction which direction is he facing now?

Assignment

- A** North
- B** South-east
- C** West
- D** South-west

[MCQ]

#Q. One day, Ravi left home and cycled 10 km Southwards, turned right and cycled 5 km and turned right and cycled 10 km and turned left and cycled 10 km. How many kilometers will he have to cycle to reach his home straight?

Assignment

- A** 5 km
- B** 15 km
- C** 12 km
- D** None of these

[MCQ]



#Q. P, Q, R and S are playing a game of carom. P,R and S,Q are partners. S is to the right of R, who is facing West. Then, Q is facing?

Assignment



North



South



East



Can't be determined

[MCQ]



#Q. One evening before sunset, two friends Sumit and Mohit were talking to each other face to face. If Mohit's shadow was exactly to his right side, which directions was Sumit facing?

Assessment

- A** East
- B** North
- C** West
- D** South

Ranking & Arrangements



3rd Left?

Right?

5th

$$L.P + R.P - 1 = \text{No. of People}$$

$$3 + 5 - 1 = 7$$

$$\text{No. of People} = 7$$

$$L.P = 3$$

$$R.P = (\text{No. of P} - L.P) + 1$$
$$= (7 - 3) + 1 = 5$$



[MCQ]



#Q. In a row of trees, one tree is the 9th from either end of the row. How many trees are there in the row?

$$\begin{aligned}\text{No. of Trees} &= L.P + R.P - 1 \\ &= 9 + 9 - 1 \\ &= 17\end{aligned}$$

- A** 16
- B** 17
- C** 18
- D** 19

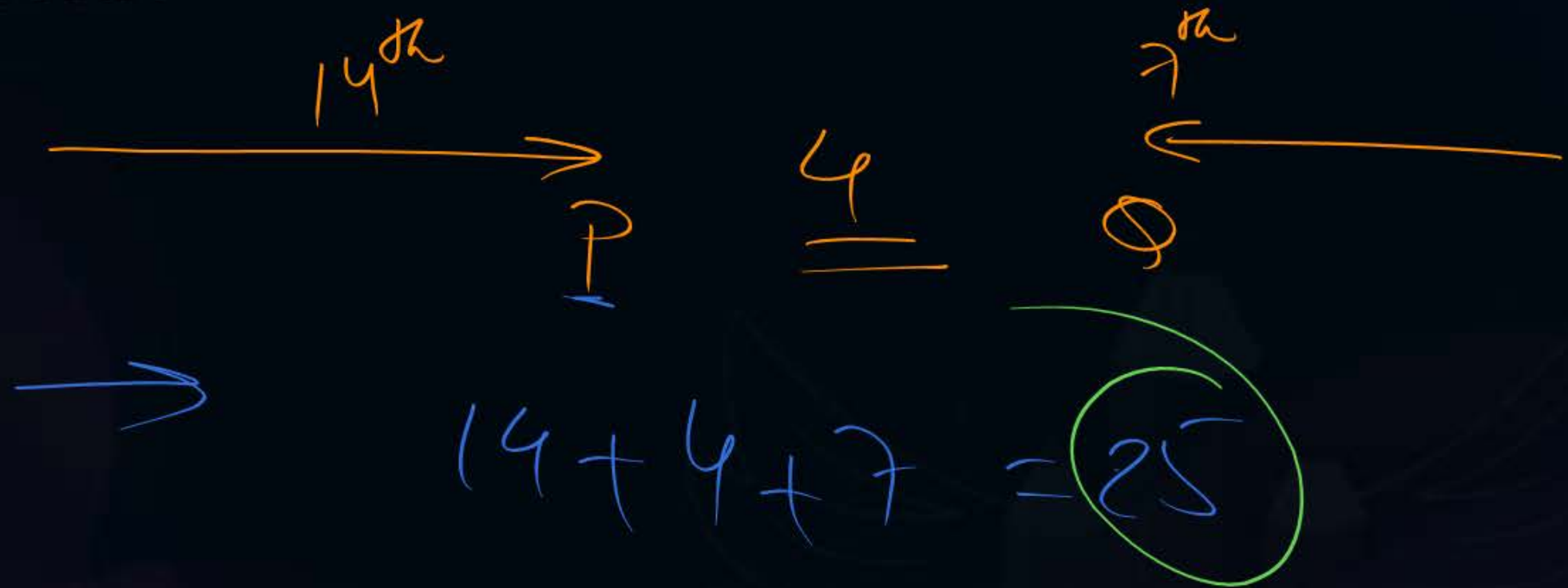


[MCQ]



#Q. Some boys are sitting in a row, P is sitting 14th from the left and Q is seventh from the right. If there are four boys between P and Q, how many boys are there in the row?

- A** 21
- B** 25
- C** 19
- D** 23



[MCQ]



#Q. In a row of girls, Kalpana who is 10th from the left and Preeti who is 9th from the right exchange their seats. If Kalpana becomes 15th from the left, then how many girls are there in a row?

A

16

B

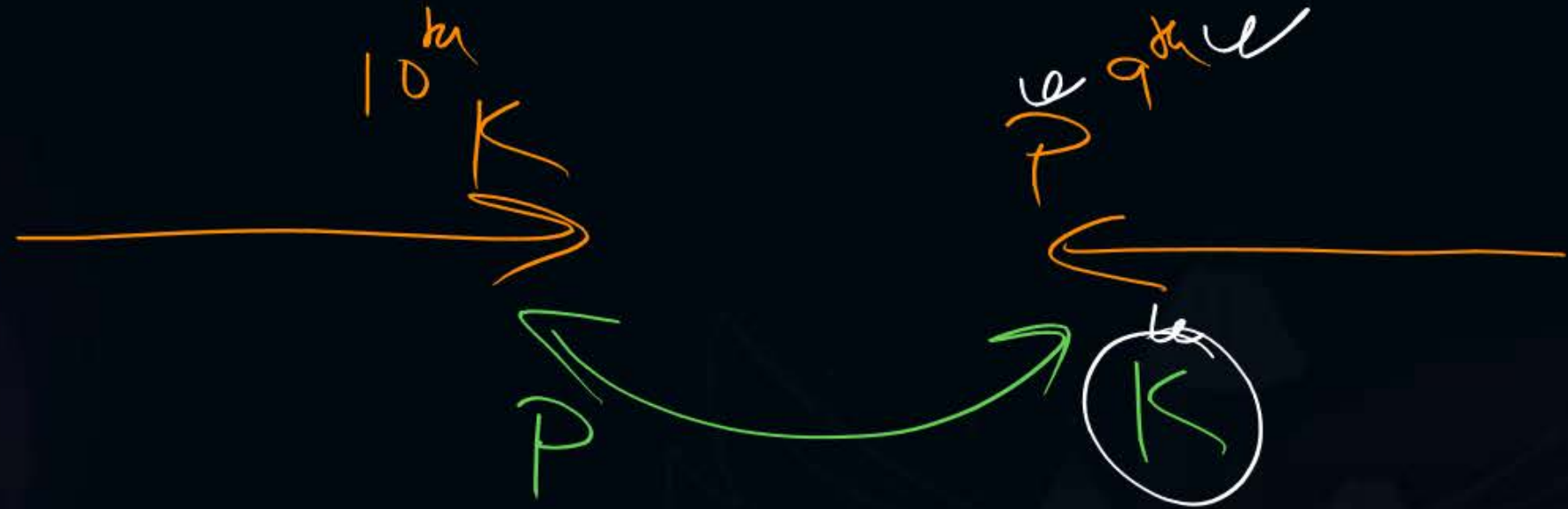
23

C

25

D

32



$$\begin{aligned} \text{No. of girls} &= 15 + 9 - 1 \\ &= 23 \end{aligned}$$



[MCQ]

$$S = 23 - 7 \quad A = 23 = 23$$



#Q. Suresh is 7 ranks ahead of Ashok in the class of 39. If Ashok's rank is 17th from the last, what is Suresh's rank from the start?

A

14

B

24

C

16

D

15

$$23 - 7 = 16^{\text{th}}$$



Ashok 17th

$$L.P = No. - R.P + 1$$

$$L.P = 39 - 17 + 1 = 23$$

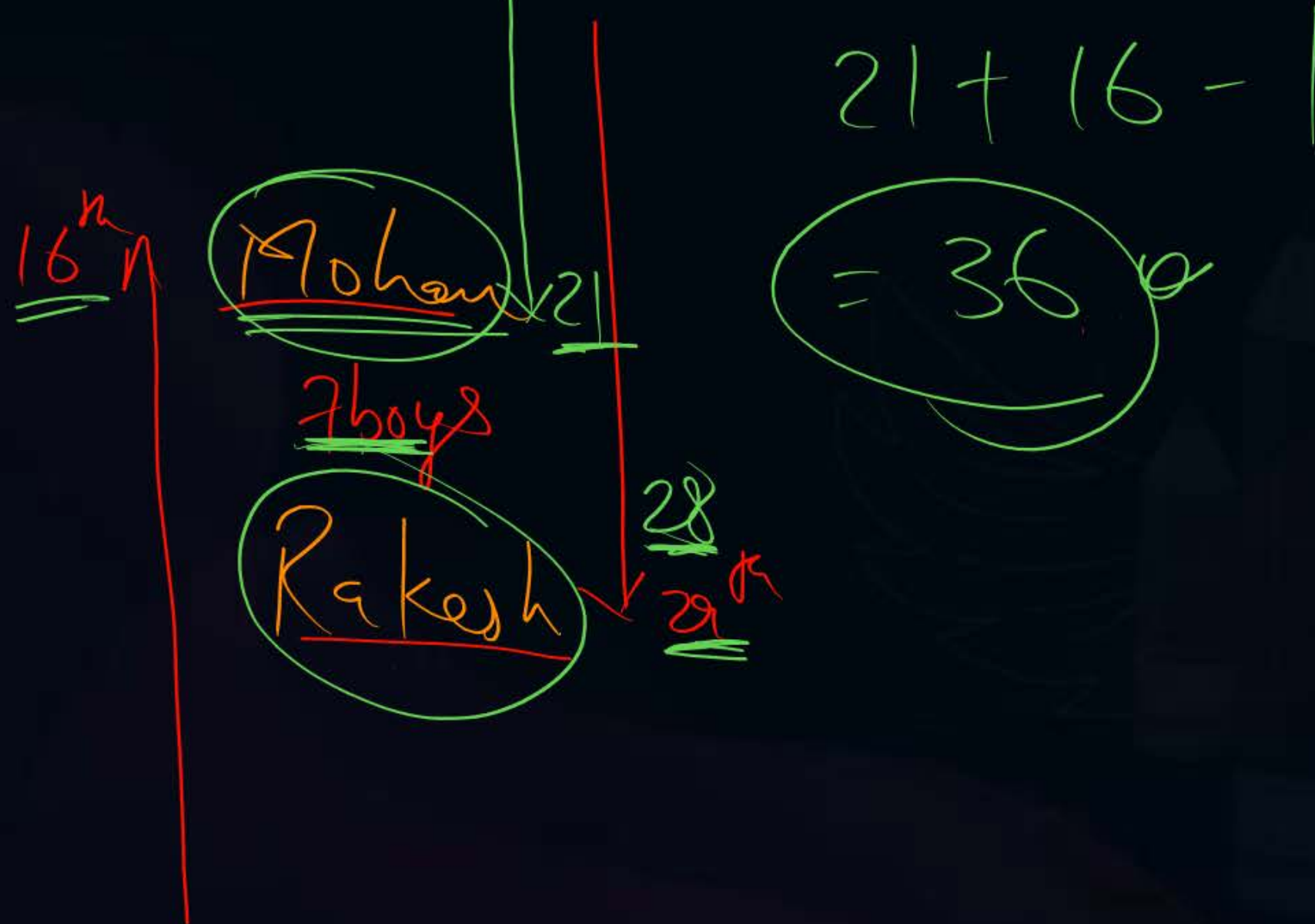


[MCQ]



#Q. In a certain class, Rakesh is 29th from the top and Mohan is 16th from the bottom in the alphabetical arrangements of names. If they have 7 boys between them what is the number of students in the class?

- A 52 ✗
- B 45
- C 36** ✓
- D 35



[MCQ]



#Q. In a row of children Moni is ninth from the left and Toni is thirteenth from the right. When they exchange places, Moni will be seventeenth from the left. Which of the following will be the new position of Toni from the right?

Assignment

- A** 20th
- B** 7th
- C** 9th
- D** 21st

[MCQ]



#Q. Rehan ranks seventh in a class of twenty. What is his rank from the last?

Assignment

- A** 13th
- B** 8th
- C** 15th
- D** 14th

[MCQ]



#Q. Of the five villages: 1) Phulwade is smaller than Dhanwade 2) Ambawade is bigger than Khelwade 3) Sonewade is bigger than Dhanwade but is not as big as Khelwade. Which is biggest village?

Amb
Khel
Sone
Dhanwade
Phul

Amb
Khel

- A** Ambawade
- B** Phulwade
- C** Dhanwade
- D** Khelwade

[MCQ]



#Q. Ashok is taller than Suresh. Raju is taller than Ashok. Chandu is shorter than Suresh, then Chandu is ____

A taller than Ashok ✗

B as tall as Suresh ✗

C taller than Suresh ✗

D shorter than Ashok ✓

Raju
Ashok
Suresh
Chandu

A handwritten diagram in orange ink shows the relative heights of four people. 'Raju' is at the top, followed by 'Ashok', then 'Suresh', and 'Chandu' at the bottom. A green curved arrow points from 'Raju' down to 'Chandu', indicating a height comparison.

[MCQ]

#Q. Four girls are swimming in a stream.

- Harjeet is further ahead of Manjula
- Neena is behind Manjula
- Ruchi is between Manjula and Neena.

Who is second from the last?

- A** Neena
- B** Manjula
- C** Harjeet
- D** Ruchi

Neena



[MCQ]



#Q. Five poles are standing in a row. M is on the left of N, O is on the right of P, which is on the right of N. If L is on the left of M, which pole is in center?

Assignment

A

L

B

M

C

N

D

O

[MCQ]



#Q. Six families A, B, C, D, E and F are living in houses in a row. B has F and D as neighbours, E has A and C as neighbours. A does not live next to D. Who are F's next door neighbours?

Assignment



B and E



B and C



B and D



Can't be determined

[MCQ]



#Q. Sudha is taller than Pushpa but shorter than Malati. Geeta is shorter than Vinu and Vinu is not as tall as Pushpa. Who should be in the middle if they stand in a row according to height?

Assignment

- A** Pushpa
- B** Malati
- C** Sudha
- D** Geeta

[MCQ]



#Q. Priti scored more than Rahul. Yamuna scored as much as Divya. Lotika scored less than Manju. Rahul scored more than Yamuna. Manju scored less than Divya. Who scored the lowest?

Assignment

- A** Yamuna
- B** Manju
- C** Lotika
- D** Rahul

[MCQ]

#Q. P, Q, R, S, and T are sitting in a row. Q is between P and T. To find out who among them is in the middle, which of the following information given in the statements A and B is/are sufficient?

Assignment

1. P is left of Q and right of S
2. R is at the right end

- A** Only {2} is sufficient
- B** Only {1} is sufficient
- C** Either {1} or {2} is sufficient
- D** Both {1} and {2} together are needed

[MCQ]



#Q. Five boys took part in a race. Ronit finished before Mohit but behind Gopal. Abbas finished before Sailesh but behind Mohit. Who won the race?

Assignment

- A** Ronit
- B** Gopal
- C** Mohit
- D** Abbas



2 mins Summary



Topic

Ranking & Arrangements

Directions



THANK - YOU